

Adaptive Latent Distribution Modeling for Industrial Time Series Anomaly Detection

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Abstract—Anomaly detection is critical for ensuring the reliability of industrial cyber-physical systems. Identifying anomalies based on data distribution is regarded as a promising approach. However, inherent noise in data collection and complex dependencies within the underlying structure can lead to class ambiguity. This ambiguity obscures the boundary between normal data and anomalies, thereby degrading the accuracy of distribution modeling. To address this issue, we shift the distribution modeling from the data space to a latent space to mitigate ambiguity and then propose a label free anomaly detection network, named ALDM. In ALDM, a contrastive-based methods is designed to facilitate the construction of a latent space, where the margin between normal data and anomalies has been expanded. Anomalies are then discerned through embeddings using a flow-based process. Recognizing the importance of distance metrics in contrastive-based methods, we propose an adaptive approach to obtain the optimal distance metric during network training instead of pre-setting a fixed formula. Furthermore, to accommodate anomalies of varying durations, we propose another event-wise performance index for evaluation. Extensive evaluations on three widely used benchmarks and a newly constructed dataset demonstrate that ALDM achieves state-of-the-art detection performance across both conventional metrics and our proposed index.

Note to Practitioners—Accurately detecting faults (anomalies) in industrial systems, such as factories or power grids, is vital to prevent costly downtime. However, noisy data and complex interactions make it challenging. Our solution, ALDM, tackles this by learning a clearer representation of system's data where normal and abnormal patterns are easier to distinguish. ALDM employs an adaptive projection to aid in anomaly detection, which minimizes expert tuning and enhances uptime. This approach improves detection accuracy, leading to enhanced system reliability and fewer unplanned stoppages. Although ALDM requires sufficient training data for reliable detection, its label-free methodology provides a practical solution for more robust monitoring in industrial systems.

Index Terms—Anomaly detection, cyber-physical systems, fault diagnosis, time series analysis, unsupervised learning.

I. INTRODUCTION

SECURITY monitoring is essential for maintaining modern infrastructures. Deep-learning-based models enhance the efficiency of monitoring and provide timely alerts for potential anomalies, thereby preventing damage from escalating. It is widely adopted in various domains, including healthcare, smart buildings, and industrial monitoring [1], [2]. Among these scenarios, industrial cyber-physical systems (CPS) are characterized by their extensive use of sensors to monitor system behavior [3]. This generates complex, multivariate dynamics, thereby making anomaly detection more challenging [4].

Traditional anomaly detection methods depend on statistical analysis, which assumes linearity, stationarity, or specific distributions [5]. These methods often struggle with high-dimensional data or complex temporal dependencies. Considering the robust capability in handling high-dimensional data, AI-based methods have been developed to improve the reliability of various industrial CPS [6].

In the scope of AI-based methods, due to the scarcity of labeled anomaly data, anomaly detection is typically studied in an unsupervised manner. Some previous methods exploit unlabeled time series data and treat anomaly detection as a one-class classification (OCC) problem. Many prediction-based methods and reconstruction-based methods are proposed to achieve anomaly detection under the OCC setting [7], [8]. The basic assumption of the OCC setting is that the training dataset covers all possible normal samples that can be easily obtained [9]. This premise is hard to hold in the real world, and its application requires additional calibration [10], [11]. Due to the typically sparse density of anomalies compared to normal instances, density-based methods have been widely explored for anomaly detection. These approaches transform the problem into estimating the likelihood that a given instance belongs to the normal data distribution. Anomalies are then detected as instances with a low probability of being normal.

Density estimation involves calculating the probability of an observed data vector. Common techniques for this task include energy-based models (EBMs), undirected probabilistic graphical models (UPGMs), autoregressive models (ARMs), and normalizing flow (NF) models [12], [13]. However, EBMs and UPGMs typically yield only approximate density estimates, limiting their suitability for anomaly detection.

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Fig. 1. Anomaly scores of existing methods on SWaT dataset. All moments are normal. Vertical drift occurs in the blue area. The moment with a higher anomaly score is more likely to be identified as an anomaly.

Similarly, ARMs suffer from computational inefficiency due to their inherently fixed sequential structure. Consequently, research in density-based anomaly detection has increasingly focused on normalizing flow (NF) models [14], which have demonstrated superior performance in industrial applications. NF models capture complex, high-dimensional data distributions by applying a learnable, invertible transformation to a simple base distribution (or prior) and leveraging the change of variables formula. Building upon this, conditional normalizing flow models [10] have been developed to exploit evolving mutual information within the data.

In industrial CPS, inherent noise and bias in data transmission and collection lead to data drifts [15], causing the distributions of normal and abnormal data to overlap. This overlap gives rise to the class ambiguity problem [16], [17], obscuring the boundary between different classes in the current data space. However, existing methods often overlook this issue and proceed to model the data distribution directly, which limits further improvement in detection performance. Figure 1 shows the anomaly detection in an industrial water treatment plant. Existing methods tend to increase the anomaly score in the blue area during data drift, leading to false alarms. To address this issue, we propose an Adaptive Latent Distribution Model (ALDM). ALDM incorporates a contrastive objective to adaptively reweight the loss function, guiding the model to learn a latent space optimized for distribution modeling. Within this space, negative data pairs are pushed apart while positive pairs are pulled closer, thereby intrinsically increasing the inter-class margin.

The distance metric plays a crucial role in the contrastive process of ALDM. While recent contrastive methods [18] use a fixed metric for assignments, they fail to enhance separability between normal and anomalous distributions. This preset metric is often suboptimal and adversely impacts learning the projection from data space to latent space. To address this, ALDM proposes a learnable, decomposition-based approach that adaptively optimizes the distance metric during training, eliminating reliance on predefined fixed formulations.

In terms of performance evaluation, anomalies in industrial CPS exhibit variable durations [19], necessitating event-level analysis for many applications. However, established metrics

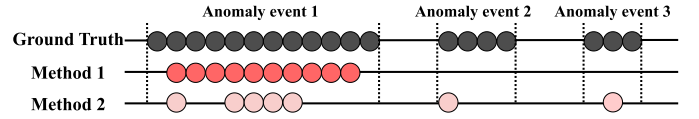


Fig. 2. Underexplored aspects of F1 score and AUROC in performance evaluation of anomaly detection.

like F1-score and AUROC focus solely on point-wise assessment, making them inadequate for event-level evaluation. Figure 2 illustrates this limitation: While Method 1 detects more anomaly points (yielding higher point-wise scores), it identifies only one anomaly event. Method 2 detects fewer points but captures all three events. To address this gap, we propose a new metric for the auxiliary of performance evaluation at the event level.

Extensive experiments are conducted on three widely used benchmarks and a newly constructed dataset. All these datasets are sourced from representative industrial CPSs. ALDM demonstrates superior detection performance, outperforming existing methods on both established point-wise metrics and our proposed event-level evaluation metric.

Compared with existing works, the contributions of this paper are summarized as follows:

- We propose ALDM, a novel unsupervised model for multivariate anomaly detection. To the best of our knowledge, this represents the first flow-based method explicitly focused on modeling latent distributions within this context.
- We design an adaptive latent distribution projection method. This innovation employs a decomposition-based learnable distance metric to construct an optimized latent space, effectively expanding the margin between normal and anomalous distributions.
- We propose a new event-level performance evaluation metric to address limitations of point-wise assessment.
- We comprehensively validate the effectiveness of ALDM on three public industrial CPS datasets and a newly constructed dataset.

The rest of this article is organized as follows. Section II reviews related work on industrial CPS anomaly detection. Section III details the proposed ALDM network. Section IV comprehensively validates the effectiveness of ALDM through extensive experiments across four datasets and three performance metrics. Finally, Section V concludes the paper.

II. RELATED WORK

A. Anomaly Detection in Industrial CPS

Industrial Cyber-Physical Systems (CPS) generate high-dimensional data with complex structures. These structures involve nonlinear dynamics and intricate spatiotemporal dependencies. Traditional anomaly detection methods [20] struggle with this complexity. They often fail to capture the necessary patterns, leading to poor detection performance. Deep neural networks provide a solution to this challenge. They excel at modeling the spatiotemporal features of multi-dimensional time series. This capability enables a shift from

shallow statistical models to deep representation learning for anomaly detection.

Labeled anomaly data is scarce in practice. Therefore, most deep learning approaches for this task are unsupervised. These methods fall into three main categories: prediction-based, reconstruction-based, and density-based approaches.

Prediction-based and reconstruction-based methods share a core idea: they learn representations of normal time series patterns. Anomalies are detected by measuring the difference between actual observations and the model's output (either predictions or reconstructions). The core module of these methods typically employs Transformer as the feature extractor to process time series data. Li et al. developed a hierarchical multi-attention block to better reconstruct normal time series [21]. Belay et al. proposed to leverage Transformer architectures and optimal truncated singular value decomposition for anomaly detection [22]. While these methods have advanced time series anomaly detection research, they face a limitation. They typically require the training data to be completely free of anomalies. This allows the model to learn an accurate representation of "normal" behavior. But in real industrial settings, ensuring perfectly clean training data is often challenging. This requirement limits the practical use of prediction and reconstruction methods.

To overcome these limitations and better use inherent data properties, researchers developed density-based methods. These approaches rely on a key observation: abnormal instances usually lie in low probability density regions compared to normal data. Among density-based techniques, Normalizing Flow (NF) stands out as particularly promising. It offers theoretical guarantees, flexible architecture, and precise density estimation.

The development of NF for industrial CPS anomaly detection has evolved over time. Dinh et al. pioneered NF for general density estimation [23]. Later, Trippe and Turner introduced Conditional Normalizing Flow (CNF) [24], enabling conditioning on relevant variables. Recognizing the dynamic nature of industrial CPS data, Zhou et al. further advanced CNF. They explicitly modeled the evolving structure of mutual dependencies [10]. Building on this progress, Li et al. explored a CNF-based approach specifically for industrial CPS anomaly detection [25]. The core principle uniting these NF-based methods is explicit. They learn a direct model of the data distribution. Anomalies are then identified as data points residing in areas of exceptionally low probability density under this learned model.

B. Distribution Modeling

Detecting anomalies by modeling the data distribution is promising. This approach has succeeded in various areas like image, audio, and time series analysis. Traditional time series distribution modeling often assumes the whole series follows a single, static distribution (e.g., Gaussian or Poisson). However, this rigid assumption hurts performance. Probabilistic models built on it often perform poorly with non-stationary sequences [9].

Modern distribution modeling uses deep generative models (DGMs). Their core idea is to model complex data distribu-

tions using neural networks [26]. DGMs are mainly grouped by how they handle the likelihood function: approximate likelihood models and exact likelihood models.

Approximate likelihood models (e.g., Variational Autoencoders (VAE) [27], Generative Adversarial Networks (GAN) [28], and Diffusion Models [29]) do not provide direct access to the exact likelihood. Instead, they optimize lower bounds or alternative objectives via variational inference and adversarial training to indirectly approximate the true data distribution.

In contrast, exact likelihood models (e.g., Normalizing Flows (NF) [30] and Autoregressive Models (ARM) [31]) can calculate the exact probability density for any data point. Their core principle involves decomposing complex distributions into simpler, tractable base distributions. This is achieved using reversible transformations or the chain rule of probability. This paradigm offers key benefits, including theoretical soundness, better training stability, and improved interpretability. Typical applications include density estimation and data compression.

Density estimation is a major task for exact likelihood models. It learns how data is generated to quantify the probability of any data point occurring. Recently, this capability has been applied to time series anomaly detection. Compared to ARM, NF avoids the sequential computation constraints. This offers NF significant computational efficiency advantages, especially with high-dimensional data. As a result, recent research has increasingly focused on using NF for time series anomaly detection [25], [32], [33].

III. METHODOLOGY

A. Overview of Anomaly Detection Framework

Consider a multivariate time series collected from an industrial CPS that includes N sensors. The time series data $\mathcal{D} = [d_1, d_2, \dots, d_L]^T \in \mathbb{R}^{L \times N}$ consists of L observations, where $d_t \in \mathbb{R}^N$ represents the system state observed at timestamp t . Note that N sensors are viewed as N variables. To prepare training samples, we partition $\mathcal{D}$ into segments using a sliding window of length T and stride S . The resulting dataset is denoted $\mathcal{X} = \{\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^M\}$, where M represents the total number of segments. We have $M = \lfloor \frac{L-T}{S} \rfloor + 1$, where $\lfloor \cdot \rfloor$ is the floor operation. The anomaly detection task is to determine whether any given segment $\mathbf{x} \in \mathcal{X}$ originating from the same CPS exhibits anomalous behavior.

To address this task, we propose the Adaptive Latent Distribution Modeling (ALDM) network, an unsupervised anomaly detection framework illustrated in Fig. 3. ALDM comprises three core modules: context learning, adaptive latent distribution projection, and anomaly determination.

The context learning module generates spatio-temporal conditioning signals to guide subsequent distribution modeling. It explicitly captures both the mutual dependencies among system variables (via self-attention) and the temporal correlations within the context (via recurrent neural networks) [10], [14]. Graph convolution then integrates these elements to produce unified conditioning features.

The Adaptive Latent Distribution Projection Module maps input segments into a latent space and estimates their probability density. This process unfolds through four sequential

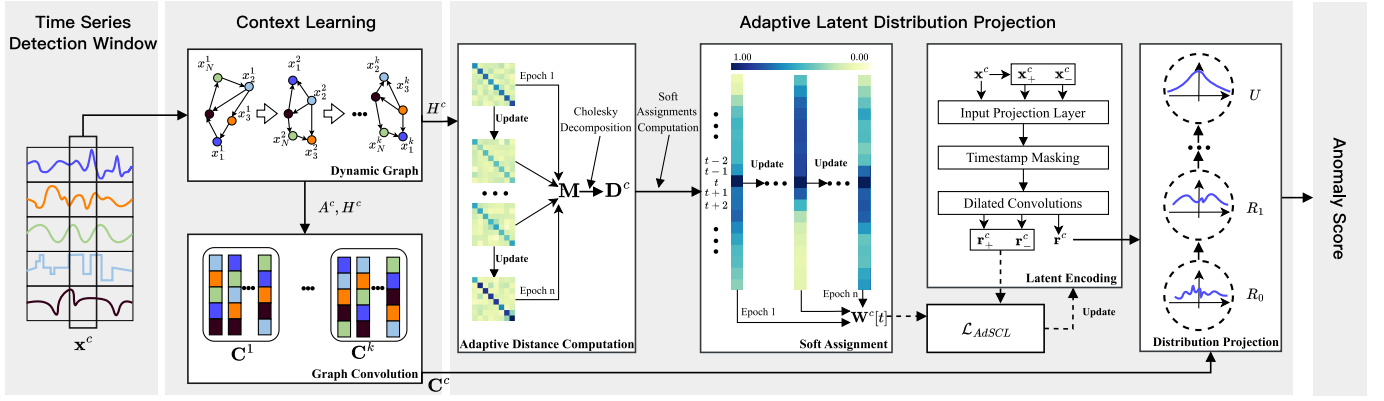


Fig. 3. Overview of ALDM. Within each detection window x^c , the time series data is first processed to generate conditional information (C^c) and temporal embeddings (H^c). This temporal embedding H^c is then used to compute a distance measure D^c . Based on D^c , soft assignments are calculated. These assignments directly form the basis for the adaptive soft contrastive loss ($\mathcal{L}_{AdSCL}$). Minimizing $\mathcal{L}_{AdSCL}$ optimizes the parameters of the latent encoding network. This optimized network transforms the input data into a latent representation r^c . Crucially, r^c becomes the explicit target for the subsequent normalizing flow to model. After the normalizing flow adaptively models the distribution of r^c , the anomaly score is finally derived from the estimated likelihood.

steps: 1. Adaptive Distance Computation: A learnable distance metric is first initialized randomly and then optimized during the training process. 2. Soft Assignments: Using this optimized metric, the module computes soft labels for each data point. 3. Latent Encoding: Guided by these soft assignments, a projection function then maps the input segments into their latent representations, denoted as $\mathbf{r}$. 4. Distribution Projection: Finally, Normalizing Flows (NFs) model the probability density of the latent vectors $\mathbf{r}$. These NFs are conditioned on the spatio-temporal features derived from the Context Learning Module. Specifically, the NFs utilize fully connected layers to parameterize invertible transformations, converting the complex distribution of $\mathbf{r}$ into a tractable base distribution, such as a Gaussian.

Finally, the anomaly determination module establishes a detection threshold using the Interquartile Range (IQR) rule applied to the estimated log-likelihoods of the latent representations. A segment x is flagged as anomalous if its log-likelihood falls below this threshold.

ALDM takes a segment x as input and outputs a binary label $\hat{y} \in \{0, 1\}$, where 0 indicates normality and 1 indicates an anomaly. The entire network trains on $\mathcal{X}$ without labels. Implementation and training specifics are detailed in the following section.

B. Context Learning

Consider data segment x^c , a multivariate time series of length T comprising N variables. The data at timestep t is denoted $x_t^c \in \mathbb{R}^N$, while the time series of the i -th variable is $x^{c,i} \in \mathbb{R}^T$. We model these N variables as nodes in a graph whose structure is learned via attention mechanisms.

For each node i , we compute its *query* and *key* vectors as:

$$q^{c,i} = x^{c,i} W^Q, \quad k^{c,i} = x^{c,i} W^K, \quad (1)$$

where $W^Q, W^K \in \mathbb{R}^{T \times T}$ are learnable weight matrices. The pairwise relationship e_{ij}^c between nodes i and j is then:

$$e_{ij}^c = \frac{(q^{c,i})(k^{c,j})^\top}{\sqrt{T}}. \quad (2)$$

The attention score a_{ij}^c (normalized influence of node i on j) and the resulting adjacency matrix A^c are:

$$a_{ij}^c = \frac{\exp(e_{ij}^c)}{\sum_{k=1}^N \exp(e_{ik}^c)}, \quad A^c = \begin{bmatrix} a_{11}^c & \cdots & a_{1N}^c \\ \vdots & \ddots & \vdots \\ a_{N1}^c & \cdots & a_{NN}^c \end{bmatrix}. \quad (3)$$

Note that this adjacency matrix represents the learned graph structure for each input segment. It is randomly initialized and subsequently refined through attention mechanisms. Specifically, the edge weight between two nodes is determined by the temporal similarity of their respective time series, computed through self-attention. This allows the model to capture context-dependent, non-linear relationships that evolve over time.

To capture temporal dynamics, we employ recurrent neural networks (RNNs; e.g., LSTM or GRU). For each timestep t in x^c , the hidden state $H_t \in \mathbb{R}^{N \times d_h}$ is updated as:

$$H_t = \text{RNN}(x_t^c, H_{t-1}), \quad (4)$$

where d_h is the hidden state dimension. The segment-level contextual embedding H^c is derived by aggregating all hidden states:

$$H^c = \text{fc}([H_1, \dots, H_T]^\top). \quad (5)$$

where fc denotes a fully connected layer.

Finally, we fuse spatial and temporal information using the learned graph structure. Given that explicitly feeding H_{t-1} into the fusion operation establishes a more direct and stable “shortcut connection” for modeling the critical short-term dynamics between adjacent steps [34], we integrating H_{t-1} in the fusion to enhance condition learning. The spatio-temporal representation C_t at time t is computed via:

$$C_t = \text{ReLU} \left(\underbrace{A^c H_t W_1}_{\text{spatial}} + \underbrace{H_{t-1} W_2}_{\text{temporal}} \right) W_3, \quad (6)$$

where the weight matrices transform features between dimensions. $W_1 \in \mathbb{R}^{d_h \times d_f}$ transforms spatial features. $W_2 \in \mathbb{R}^{d_h \times d_f}$ transforms temporal features. $W_3 \in \mathbb{R}^{d_f \times d_o}$ enhances representational capacity. Note that d_f is the fused feature dimension

and d_o is the output dimension. The full spatio-temporal embedding is $\mathbf{C}^c = [\mathbf{C}_1, \dots, \mathbf{C}_T]^\top \in \mathbb{R}^{T \times N \times d_o}$.

C. Adaptive Latent Distribution Projection

The adaptive latent distribution projection module consists of four components: adaptive distance computation, soft assignment, latent encoding, and distribution projection. In this module, adaptive distance computation and soft assignment collaboratively guide the latent encoding network in mapping input data to representations within a latent space. Subsequently, the distribution projection component builds a transformation chain that maps complex latent distributions onto simple base distributions. We formalize this process below.

1) *Adaptive Distance Computation*: We define a positive semi-definite distance metric matrix $\mathbf{M}$ via Cholesky decomposition:

$$\mathbf{M} = \mathbf{L}^\top \mathbf{\Lambda} \mathbf{L}, \quad (7)$$

where $\mathbf{L} \in \mathbb{R}^{E \times E}$ is a lower triangular matrix and $\mathbf{\Lambda} \in \mathbb{R}^{E \times E}$ is a diagonal matrix. Both $\mathbf{L}$ and $\mathbf{\Lambda}$ are initialized randomly and optimized using gradient descent during training. This decomposition ensures numerical stability and mitigates overfitting compared to direct learning of $\mathbf{M}$.

Unlike prior approaches that compute distances directly from raw data, we propose to compute contextual distance between temporal embeddings. Let $\mathbf{H}^c = [\mathbf{h}_1, \dots, \mathbf{h}_T]^\top \in \mathbb{R}^{T \times E}$ denote the temporal embeddings from RNN in context learning. The distance matrix $\mathbf{D}^c \in \mathbb{R}^{T \times T}$ is computed as:

$$\mathbf{D}^c[t][t'] = \sqrt{(\mathbf{h}_t - \mathbf{h}_{t'})^\top \mathbf{M} (\mathbf{h}_t - \mathbf{h}_{t'})} \quad (8)$$

$$= \sqrt{(\mathbf{h}_t - \mathbf{h}_{t'})^\top \mathbf{L}^\top \mathbf{\Lambda} \mathbf{L} (\mathbf{h}_t - \mathbf{h}_{t'})}, \quad (9)$$

where $\mathbf{h}_t, \mathbf{h}_{t'} \in \mathbb{R}^E$ are temporal embeddings at timesteps t and t' . Equation (8) represents the Mahalanobis distance, which generalizes Euclidean distance (obtained when $\mathbf{M} = \mathbf{I}$) by incorporating covariance structure. This learnable formulation adapts to data distributions, overcoming limitations of preset metrics in prior work.

2) *Soft Assignment*: From $\mathbf{D}^c$, we compute a soft assignment matrix $\mathbf{W}^c \in \mathbb{R}^{T \times T}$:

$$\mathbf{W}^c[t][t'] = 2\sigma(-\tau_T \cdot \mathbf{D}^c[t][t']), \quad (10)$$

where $\tau_T > 0$ controls distribution sharpness and σ is the sigmoid function. Soft assignments provide adaptive similarity measures that capture the correlation of temporal embeddings, enhancing the flexibility of the contrastive process.

3) *Latent Encoding*: Given a batch of K time series segments $\mathcal{X} = \{\mathbf{x}^1, \dots, \mathbf{x}^K\}$, where each $\mathbf{x}^c \in \mathbb{R}^{T \times N}$ ($c \in [1, K]$), we generate augmented views as follows: For each segment, we sample two overlapping intervals $[a_1, b_1]$ and $[a_2, b_2]$ satisfying $0 < a_1 \leq a_2 \leq b_1 \leq b_2 \leq T$. The augmented views $\mathbf{x}_+^c$ and $\mathbf{x}_-^c$ are created by masking all elements outside $[a_1, b_1]$ and $[a_2, b_2]$, respectively. This yields augmented pairs $\{(\mathbf{x}_+^1, \mathbf{x}_-^1), \dots, (\mathbf{x}_+^K, \mathbf{x}_-^K)\}$.

A feature extractor f_θ^{AdSC} projects segments to latent representations:

$$\mathbf{r}^c = f_\theta^{\text{AdSC}}(\mathbf{x}^c) \in \mathbb{R}^{T \times N}, \quad (11)$$

where θ denotes trainable parameters. The encoder f_θ^{AdSC} consists of three components, including an input projection layer, a timestamp masking module, and a dilated convolutional neural network (CNN) module.

Specifically, for each input segment $\mathbf{x}^c \in \mathbb{R}^{T \times N}$, the processing pipeline is as follows:

- 1) *Input Projection*: A fully connected layer maps each timestamp observation $\mathbf{x}^{c,t} \in \mathbb{R}^N$ to a high-dimensional latent vector $\mathbf{r}^{c,t} \in \mathbb{R}^d$:

$$\mathbf{r}^{c,t} = \text{fc}(\mathbf{x}^{c,t})$$

- 2) *Timestamp Masking*: This module randomly selects timestamps and masks their corresponding vectors to generate augmented context views. Note that the masking operates on latent representations rather than raw temporal indices. We choose to mask latent vectors instead of raw values. This decision is based on the fact that time series data can be unbounded ($\mathbb{R}$), and thus lacks a natural masking token.
- 3) *Dilated CNN*: A 10-block residual network processes the sequence $\{\mathbf{r}^{c,1}, \dots, \mathbf{r}^{c,T}\}$. Each block contains two 1-D convolutional layers with exponentially increasing dilation rates: for the l -th block ($l \in \{1, \dots, 10\}$), the dilation parameter is 2^l .

The latent encoding network enhances separability between normal and anomalous patterns. The encoder parameters θ are optimized through a composite contrastive loss function containing two complementary components:

- (1) *Instance-level loss promotes segment discrimination*: Let c be the index of the input time series segment, t be the timestamp, $\mathbf{r}_+^{c,t}$ and $\mathbf{r}_-^{c,t}$ be the representation of $\mathbf{x}_+^c$ and $\mathbf{x}_-^c$ for the timestamp t , $\mathcal{L}_I^{(c,t)}$ is denoted as

$$\mathcal{L}_I^{(c,t)} = -\log \frac{\exp(\mathbf{r}_-^{c,t} \cdot \mathbf{r}_+^{c,t})}{\sum_{j=1}^B \left[\exp(\mathbf{r}_-^{c,t} \cdot \mathbf{r}_+^{j,t}) + \mathbb{I}_{[c \neq j]} \exp(\mathbf{r}_-^{c,t} \cdot \mathbf{r}_-^{j,t}) \right]} \quad (12)$$

where B is the batch size, and $\mathbb{I}$ is the indicator function. $\mathcal{L}_I$ measures the similarity of segments, guiding the model to learn robust representations through instance-wise contrast.

- (2) *Temporal loss captures contextual relationships*:

$$\mathcal{L}_T^{(c,t)} = -\log p_c^+(t, t') - \text{AdL}(c, t) \quad (13)$$

where

$$\begin{aligned} \text{AdL}(c, t) &= \sum_{t'=1, t' \neq t}^T W_T(t, t') (\log p_c^+(t, t') + \log p_c^-(t, t')) \\ p_c^+(t, t') &= \frac{\exp(\mathbf{r}_-^{c,t} \cdot \mathbf{r}_+^{c,t'})}{\sum_{s \in \Omega} (\exp(\mathbf{r}_-^{c,t} \cdot \mathbf{r}_+^{c,s}) + \mathbb{I}_{[s \neq t]} \exp(\mathbf{r}_-^{c,t} \cdot \mathbf{r}_-^{c,s}))} \\ p_c^-(t, t') &= \frac{\exp(\mathbf{r}_-^{c,t} \cdot \mathbf{r}_-^{c,t'})}{\sum_{s \in \Omega} (\exp(\mathbf{r}_-^{c,t} \cdot \mathbf{r}_+^{c,s}) + \mathbb{I}_{[s \neq t]} \exp(\mathbf{r}_-^{c,t} \cdot \mathbf{r}_-^{c,s}))} \end{aligned}$$

Note that Ω denotes timestamps in the augmented views' overlap region.

The composite contrastive loss is:

$$\mathcal{L}_{\text{AdSCL}} = \frac{1}{KT} \sum_{c=1}^K \sum_{t=1}^T \left(\mathcal{L}_T^{(c,t)} + \mathcal{L}_1^{(c,t)} \right). \quad (14)$$

where K is the number of instances, T is the sequence length.

4) *Distribution Projection*: This component utilizes normalizing flows to transform the complex high-dimensional data distribution $p_X(\mathbf{x})$ into a simple base distribution $\pi_U(\mathbf{u})$ through a series of learnable, invertible affine transformations. Following established approaches [34], we incorporate spatio-temporal correlations as conditioning information to enhance density estimation. This is formalized through a conditional normalizing flow:

$$\mathbf{u} = f_\theta(\mathbf{x} | \mathbf{C}) \quad (15)$$

where $\mathbf{C}$ represents the spatio-temporal conditioning variable, $f_\theta: \mathcal{X} \rightarrow \mathcal{U}$ is a parametric, bijective transformation. θ denotes the parameters of f_θ , and $\mathbf{u} \sim \pi_U(\mathbf{u})$ with $\pi_U(\mathbf{u})$ typically chosen as a standard Gaussian distribution.

Unlike previous work that estimates density in the data space ($p_X(\mathbf{x})$), we propose modeling the density of learned representations $\mathbf{r} = f_\theta^{\text{AdSC}}(\mathbf{x})$ in latent space. To our knowledge, this represents the first flow-based method explicitly focused on modeling latent distributions within this context.

The formal density estimation process is as follows: Given a segment $\mathbf{x}^c \sim p_X(\mathbf{x})$ and its associated condition $\mathbf{C}^c$, we first obtain the latent representation $\mathbf{r}^c = f_\theta^{\text{AdSC}}(\mathbf{x}^c)$. The density of $\mathbf{r}^c$ is then given by:

$$p_R(\mathbf{r}^c | \mathbf{C}^c) = \pi_U(f_\theta(\mathbf{r}^c | \mathbf{C}^c)) \left| \det \frac{\partial f_\theta(\mathbf{r}^c | \mathbf{C}^c)}{\partial \mathbf{r}^c} \right| \quad (16)$$

where $\pi_U(\cdot) = \mathcal{N}(\boldsymbol{\mu}, \mathbf{I})$ and the Jacobian determinant $\left| \frac{\partial f_\theta}{\partial \mathbf{r}} \right|$ accounts for the volume change under the transformation.

During training, the conditional normalizing flow (CNF) is optimized jointly with the context learning module via maximum likelihood estimation. The objective function is:

$$\begin{aligned} \mathcal{L}_{\text{MLE}} &= -\log p_R(\mathbf{r}^c | \mathbf{C}^c) \\ &= -\frac{1}{NK} \sum_{c=1}^K \sum_{n=1}^N \log \left[\pi_U(f_\theta(\mathbf{r}_n^c | \mathbf{C}_n^c)) \left| \det \frac{\partial f_\theta}{\partial \mathbf{r}_n^c} \right| \right] \\ &\approx \frac{1}{NK} \sum_{c=1}^K \sum_{n=1}^N \left[\frac{1}{2} \|\hat{\mathbf{u}}_n^c - \boldsymbol{\mu}_n\|_2^2 + \log \left| \det \frac{\partial f_\theta}{\partial \mathbf{r}_n^c} \right| \right] \end{aligned} \quad (17)$$

This MLE objective $\mathcal{L}_{\text{MLE}}$ will integrate with contrastive loss $\mathcal{L}_{\text{AdSCL}}$ to form the complete optimization target.

D. Anomaly Determination

An anomaly is flagged when the representation vector $\mathbf{r}^c$ falls within a low-density region of the learned distribution. This position indicates that the original input $\mathbf{x}^c$ is more likely to be anomalous. The anomaly score S^c for c -th sample is computed as the mean negative log-likelihood across all N dimensions of its embedding $\mathbf{r}^c$:

$$S^c = -\frac{1}{N} \sum_{n=1}^N \log p_R(\mathbf{r}^{c,n}) \quad (18)$$

Algorithm 1 Training Procedure for ALDM

Input: Training Data: $\mathcal{D}_{\text{train}} = \{\mathbf{x}^i\}_{i=1}^{N_1}$; Learning Rate: α ; Total Epochs: E ; Batch Size: B ; Loss Weight: λ ; Networks: $f_{\theta_1}^{\text{CL}}, f_{\theta_2}^{\text{AdSC}}, f_{\theta_3}^{\text{NF}}$.

Output: Trained model parameters $\theta_1, \theta_2, \theta_3$.

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1: epoch  $\leftarrow$  0                                ▶Initialize epoch counter
2: while epoch <  $E$  do
3:   idx1  $\leftarrow$  0                                ▶Reset batch index for each epoch
4:   while idx1 <  $\lceil N_1/B \rceil$  do                ▶Iterate over batches
5:      $\mathbf{x}^c \leftarrow \text{SampleBatch}(\mathcal{D}_{\text{train}}, \text{idx}_1, B)$  ▶Sample mini-batch  $\mathbf{x}^c$  of size  $B$ 
6:      $\mathbf{C}^c \leftarrow f_{\theta_1}^{\text{CL}}(\mathbf{x}^c)$  ▶Compute spatio-temporal condition  $\mathbf{C}^c$ 
7:      $\mathbf{r}^c, \mathbf{W}(t, t') \leftarrow f_{\theta_2}^{\text{AdSC}}(\mathbf{x}^c)$  ▶Learn latent embeddings  $\mathbf{r}^c$  and adaptive assignments  $\mathbf{W}$ 
8:      $\hat{\mathbf{u}}^c \leftarrow f_{\theta_3}^{\text{NF}}(\mathbf{r}^c, \mathbf{C}^c)$  ▶Estimate latent density
9:      $\mathcal{L}_{\text{AdSCL}} \leftarrow \mathcal{L}_{\text{AdSCL}}(\mathbf{r}^c, \mathbf{W}(t, t'))$  ▶Compute adaptive soft contrastive loss
10:     $\mathcal{L}_{\text{MLE}} \leftarrow \mathcal{L}_{\text{MLE}}(\hat{\mathbf{u}}^c, \mathbf{r}^c)$  ▶Compute maximum likelihood estimation loss
11:     $\mathcal{L}_{\text{ALDM}} \leftarrow (1 - \lambda) \cdot \mathcal{L}_{\text{MLE}} + \lambda \cdot \mathcal{L}_{\text{AdSCL}}$  ▶Compute combined loss
12:     $\theta_1 \leftarrow \theta_1 - \alpha \cdot \nabla_{\theta_1} \mathcal{L}_{\text{ALDM}}$  ▶Update GL parameters
13:     $\theta_2 \leftarrow \theta_2 - \alpha \cdot \nabla_{\theta_2} \mathcal{L}_{\text{ALDM}}$  ▶Update AdSCL parameters
14:     $\theta_3 \leftarrow \theta_3 - \alpha \cdot \nabla_{\theta_3} \mathcal{L}_{\text{ALDM}}$  ▶Update NF parameters
15:    idx1  $\leftarrow$  idx1 + 1                            ▶Increment batch index
16:  end while
17:  epoch  $\leftarrow$  epoch + 1                            ▶Increment epoch counter
18: end while

```

where $\mathbf{r}^{c,n}$ denotes the n -th dimension of the embedding vector $\mathbf{r}^c$.

Directly thresholding the anomaly score S^c to label anomalies has been shown to increase anomaly detection interference [35]. To mitigate this, we compute the anomaly scores S^c for the entire training set. The detection threshold is then robustly set using the Interquartile Range (IQR) method:

$$\text{Threshold} = Q_3 + 1.5 \times (Q_3 - Q_1) \quad (19)$$

where Q_1 and Q_3 represent the 25th and 75th percentiles of the anomaly scores S^c observed in the training data.

E. Joint Optimization

The proposed model employs joint optimization of the context learning module and the adaptive latent distribution projection network to enhance training stability. The overall objective function for ALDM is formulated as:

$$\mathcal{L}_{\text{ALDM}} = (1 - \lambda) \cdot \mathcal{L}_{\text{MLE}} + \lambda \cdot \mathcal{L}_{\text{AdSCL}} \quad (20)$$

where $\lambda \in [0, 1]$ is a hyperparameter balancing the contribution of each loss component ($\lambda = 0.5$ by default). The loss terms $\mathcal{L}_{\text{AdSCL}}$ and $\mathcal{L}_{\text{MLE}}$ are defined by Equations (14) and (17), respectively.

Algorithm 1 details the ALDM training procedure.

IV. EXPERIMENTS AND ANALYSIS

This section first introduces the proposed event-level auxiliary metric and a new anomaly detection dataset. We then

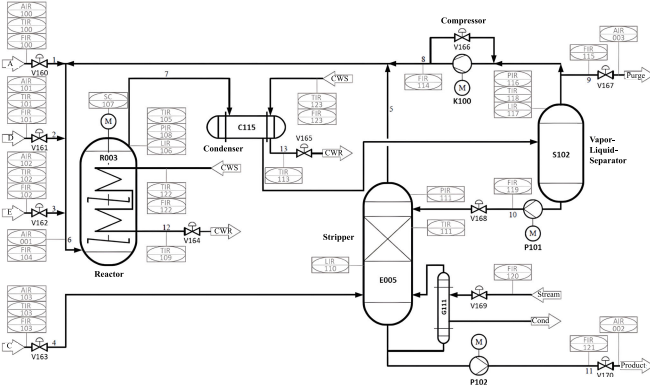


Fig. 4. Piping and instrumentation diagram of the TEPE.

detail the experimental setup, present comparative results across four datasets using three evaluation metrics, and conduct ablation studies to validate the effectiveness of the proposed modules.¹

A. Auxiliary Metric and New Dataset

1) *Event-Level Auxiliary Metric*: As shown in Fig. 2, classic metrics exhibit limitations in capturing event-level anomaly characteristics. To address this, we propose an auxiliary metric based on anomaly event recall. Adopting the standard time series definition, an *anomaly event* is a contiguous sequence of anomalous points, while a *prediction segment* is a contiguous set of points flagged as anomalous by the model.

We define the true positive predictions TP_E as the number of ground truth (GT) events that fully or partially overlap with one prediction segment or more. The false negative predictions FN_E is the number of GT events that do not overlap with any prediction segment. Event recall (R_E) is computed as:

$$R_E = \frac{TP_E}{TP_E + FN_E} \quad (21)$$

Event precision (P_E) incorporates point-level false positives with a normalization factor:

$$P_E = \frac{TP}{TP + FP} \times \lambda_{FE}, \quad \lambda_{FE} = 1 - \frac{FP}{N} \quad (22)$$

where TP/FP denote point-level true/false positives, and N is the total normal points. The event-level F1 score is:

$$F1_E = \frac{2 \cdot P_E \cdot R_E}{P_E + R_E} \quad (23)$$

Alongside $F1_E$, we report classic point-level F1 score and AUROC for comprehensive evaluation.

2) *TEPE Simulation Dataset*: As illustrated in Fig. 2, we would like to perform an event-level analysis for anomaly detection. However, existing public datasets contain a relatively small number of anomaly events (e.g., 35 in SWaT, 72 in PSM, 36 in MSL). To overcome this limitation, we developed a new simulation dataset TEPE (Tennessee Eastman Process Extension), extending the benchmark Tennessee Eastman Process (TEP) [36] to include diverse anomaly events.

The base TEP models a plant-wide chemical process featuring five major unit operations: reactor, product condenser, vapor-liquid separator, recycle compressor, and product stripper. The TEPE dataset contains 53 process variables, including flow rates, pressures, temperatures, levels, and chemical component concentrations. It incorporates 20 distinct anomaly types, simulating 180 artificial anomaly events at 1-minute sampling intervals over 360 operational days. Fig. 4 shows the piping and instrumentation diagram.

B. Experimental Setup

1) *Datasets and Baselines*: Apart from the proposed new dataset TEPE, three additional datasets are evaluated:

- **SWaT (Secure Water Treatment)** [37]: Originating from a scaled-down version of an industrial water treatment plant, the SWaT dataset is used for research in cybersecurity and anomaly detection in critical infrastructure systems. SWaT collects 51 sensor data from a real-world industrial water treatment plant, at a frequency of one second. The dataset provides ground truths of 41 attacks launched during 4 days.
- **MSL (Mars Science Laboratory rover)** [38]: This dataset originates from the Mars Science Laboratory rover, specifically the Curiosity rover. It includes telemetry data used to detect anomalies in the rover's operational parameters during its mission on Mars.
- **PSM (Pooled Server Metrics)** [39]: This dataset aggregates performance metrics from multiple server nodes managed by eBay. It is utilized to identify outliers, indicating potential issues in server performance or security breaches.

For the baseline methods, we used publicly available implementations under the contaminated training setting. These methods include DeepSAD [40], DeepSVDD [41], USAD [42], DAGMM [43], GANF [34], MTGFlow [10], FITS [44], and CrossAD [45]. We retained their original network and trained them follows the best hyperparameter reported in their paper.

2) *Implementation Details*: All experiments were conducted on an Ubuntu Linux 64-bit operating system, with the support of Python 3.8 environment and Pytorch platform. The hardware configuration is listed as follows: an Intel Core i9-13900KF CPU @5.8 GHz, a 32 GB memory card, and an NVIDIA RTX 4090 @24 GB GPU.

In the comparison study, we set the anomaly detection window size to 60 and the stride size to 10 for all experiments. The total number of training epochs is 40 (TEPE), 100 (SWaT), 250 (PSM), and 250 (MSL). For the batch size, we used 512 for the SWaT dataset and 256 for the other datasets. Adaptive Moment Estimation with Weight Decay (AdamW) is chosen as the optimizer, and the learning rate scheduler is OneCycleLR with a percentage start of 0.2, an initial learning rate of 0.002, and a maximum learning rate of 0.004. Detailed parameter settings of our model are provided in Table I.

For the parameter selection of our model, we aligned the general parameters such as window size, stride size, weight decay, batch size, threshold ratio, and block size with those

¹The source code of ALDM and the proposed TEPE dataset are available at <https://github.com/YuLiu-61/ALDM>

TABLE I
EXPERIMENTAL PARAMETERS OF ALDM IN COMPARATIVE EXPERIMENTS

Dataset	Window size	Window stride	Weight Decay	Initial/maximum learning rate	Start percentage	Anomaly ratio	Batch size	Epoches	Threshold ratio
SWaT	60	10	5×10^{-4}	$2 \times 10^{-3}/4 \times 10^{-3}$	0.2	0.121	512	100	$Q_3 + 1.5 \times IQR$
PSM						0.278	256	250	
MSL						0.105	256	250	
TEPE						0.208	256	40	

TABLE II
EXPERIMENTAL RESULTS OF ALDM AND BENCHMARK METHODS ON AUROC(%)

Model	SWaT	PSM	MSL	TEPE
DeepSVDD	66.8±2.0	67.5±1.4	60.8±0.4	76.0 ±1.6
DeepSAD	75.4±2.4	73.2±3.3	61.6±0.6	65.1±4.2
USAD	78.8±1.0	78.0±0.2	57.0±0.1	<u>81.6 ±0.2</u>
DAGMM	72.8±3.0	64.6±2.6	56.5±2.6	50.5±3.9
GANF	79.8±0.7	81.8±1.5	64.5±1.9	80.8±0.5
FITS	80.1±1.1	81.5±0.6	60.4±0.7	69.1±0.8
MTGFlow	<u>84.8±1.5</u>	<u>85.7±1.5</u>	<u>67.2±1.7</u>	81.1±0.2
CrossAD	70.9±0.8	76.9±1.2	63.4±0.1	72.4±0.1
ALDM [Ours]	90.5±1.3	86.8±3.4	70.1±0.8	82.5±0.8

of the open-sourced state-of-the-art density-based method MTGFlow, as detailed in Table I. With these parameters set, we further optimized the remaining tunable hyperparameters through extensive experimentation.

C. Performance Evaluation of ALDM

The results across the four datasets on three metrics are presented in Tables II and III. Bold and underlined numbers denote the optimal and suboptimal values, respectively.

As shown in Table II, the ALDM model proposed in this paper exhibits the best AUROC performance across all scenarios, including cyber-physical attacks in SWaT and PSM, as well as system-level faults in TEPE and MSL. Notably, on the SWaT dataset, the AUROC of ALDM reaches 0.905, marking a 5.7 percentage point improvement over the suboptimal model MTGFlow. MTGFlow ranks second overall, with its AUROC value (0.811) on the TEPE dataset slightly lower than the USAD model by 0.5 percentage points. In terms of standard deviations, ALDM shows stable performance on datasets with low anomaly ratios but relatively higher variance on datasets with more anomalies. This instability stems from its density-based framework and latent encoding, which may inadvertently incorporate prevalent anomalous patterns into the learned latent structure, biasing the likelihood estimation and affecting the detection threshold. In practice, this effect can be effectively mitigated by collecting more normal data.

Table III presents a comparison on F1 and $F1_E$ scores. The proposed ALDM persists the superior F1 score across all datasets, with a particularly notable performance on the SWaT, where it outperforms the second-ranked model by 0.12 (0.68 vs 0.56). The $F1_E$ score indicates that ALDM still delivers the best performance. The suboptimal $F1_E$ for SWaT, PSM, MSL, and TEPE are achieved by DAGMM (0.7734), MTGFlow (0.6743), MTGFlow (0.4949), and DeepSVDD

(0.7403), respectively. It is noteworthy that the suboptimal model on the $F1_E$ score significantly differs from the point-level indicator (F1/AUROC). For example, DeepSVDD ranks second in the $F1_E$ of TEPE, but its F1 score is lower than that of MTGFlow. This verifies the difference between event-level indicators and point-level indicators: the former requires the model to have the ability to identify the temporal continuity of abnormal events, while the latter focuses on the detection accuracy of individual points. This finding confirms the supplementary value of the $F1_E$ score to the existing evaluation metrics.

To visualize the benefits of ALDM, Fig. 5 presents the ROC curves for four datasets. In these curves, ALDM consistently outperforms the baseline methods, highlighting its robustness and effectiveness in anomaly detection tasks.

Fig. 6 uses a point plot to compare the anomaly predictions of the proposed ALDM and the second-ranked MTGFlow. Time indices are on the x-axis, and anomaly scores per data segment are on the y-axis. Red backgrounds highlight ground-truth anomalies. ALDM consistently assigns higher scores to anomalies (within red regions) and lower scores to normal data compared to MTGFlow. It also exhibits sharper score changes at anomaly boundaries (beginnings/ends of red region), indicating more precise boundary detection. Besides, ALDM achieves a significantly lower mean anomaly score for normal periods, demonstrating superior false alarm suppression.

D. Result Verification

Recall that the core problem comes from the observation that inherent noise and bias in data transmission and collection induce data drifts. These drifts cause the distributions of normal and abnormal data to overlap. The overlap gives rise to the class ambiguity problem, where the boundary between classes becomes obscured in the current data space. In the following, we evaluate the effectiveness of ALDM in addressing the above issue.

We first employ the Augmented Dickey-Fuller (ADF) test statistics [46] to measure the extent of data drifts. The results in Table IV reveal clear differences in time series stationarity across the four datasets. SWaT (ADF = -0.780, $p = 0.83$) and MSL (ADF = -1.228, $p = 0.66$) show high p -values, indicating strong non-stationarity and significant data drift. In contrast, TEPE (ADF = -9.48, $p = 0.01$) is clearly stationary, while PSM (ADF = -6.060, $p = 0.05$) sits near the conventional significance threshold, suggesting mild or borderline non-stationarity.

TABLE III
EXPERIMENTAL RESULTS OF ALDM AND BENCHMARK METHODS ON $F1_E$ AND $F1$ SCORE. POINT ADJUSTMENT NOT APPLIED

	SWaT			PSM			MSL			TEPE		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
DeepSAD	0.3800	0.6107	0.4685	0.6352	0.7597	0.6919	0.5747	0.2336	0.3322	0.3977	0.3226	0.3563
DeepSVDD	0.3363	0.6336	0.4394	0.6217	0.6939	0.6558	0.7042	0.2336	0.3509	0.6621	0.5280	0.5875
USAD	0.3264	0.3893	0.3551	0.4767	0.9202	0.6280	0.8261	0.1484	0.2517	0.6148	0.5940	0.6042
DAGMM	0.9304	0.4084	<u>0.5676</u>	0.5868	0.7006	0.6387	0.1019	0.4159	0.1638	0.1313	0.4207	0.2001
GANF	0.2743	0.6355	0.3832	0.6355	0.8249	0.7179	0.7200	0.2533	0.3737	0.5797	0.5701	0.5749
FITS	0.2359	0.0774	0.1165	1	0.0888	0.1630	0.6372	0.1101	0.1876	0.9315	0.3968	0.5566
MTGFlow	0.3524	0.7290	0.4751	0.6957	0.8181	<u>0.7520</u>	0.9608	0.2290	<u>0.3698</u>	0.6231	0.5894	<u>0.6057</u>
CrossAD	0.1994	0.4333	0.2731	0.5805	0.8553	0.6979	0.9452	0.2584	0.4058	0.6311	0.4644	0.5351
ALDM [Ours]	0.8223	0.5916	0.6881	0.7661	0.8211	0.7926	0.9464	0.2477	0.3926	0.6814	0.5646	0.6175

	SWaT			PSM			MSL			TEPE		
	P_E	R_E	$F1_E$	P_E	R_E	$F1_E$	P_E	R_E	$F1_E$	P_E	R_E	$F1_E$
DeepSAD	0.3928	1	0.5436	0.6352	0.8421	0.5989	0.5747	0.4	0.4691	0.4732	0.9444	0.6110
DeepSVDD	0.4851	1	0.6360	0.6410	0.6842	0.5538	0.8571	0.3333	0.4796	0.6305	0.9444	<u>0.7403</u>
USAD	0.3398	0.8889	0.4757	0.4767	0.8	0.3604	0.8261	0.3	0.4398	0.6148	0.7778	0.6696
DAGMM	0.9227	0.6667	<u>0.7734</u>	0.5868	0.7895	0.5387	0.1019	0.7333	0.1322	0.1313	1	0.1623
GANF	0.2893	1	0.4188	0.6355	0.8421	0.5871	0.1645	0.5333	0.2279	0.5797	0.7222	0.6254
FITS	0.0734	0.7200	0.1332	1	0.5151	<u>0.6800</u>	0.0728	0.3	0.1173	0.3944	0.5414	0.5456
MTGFlow	0.6861	0.8889	0.7702	0.7350	0.7368	0.6743	0.9608	0.3333	<u>0.4949</u>	0.6231	0.7778	0.6755
CrossAD	0.0220	0.7500	0.0427	0.7772	0.3529	0.4854	0.0123	0.5714	0.0240	0.1293	0.5862	0.2119
ALDM [Ours]	0.8223	1	0.8985	0.7661	0.8947	0.7516	0.7903	0.4	0.5303	0.6954	0.8333	0.7460

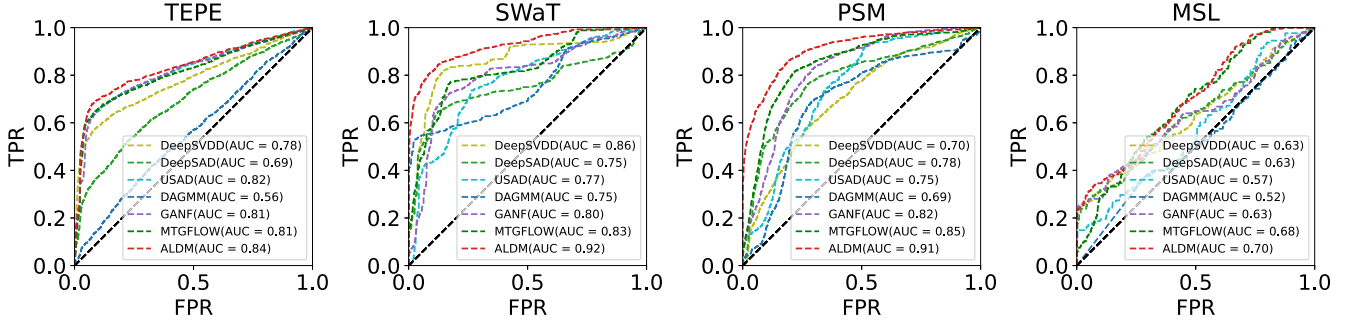


Fig. 5. The AUROC curves of the SWaT, PSM, MSL, and TEPE datasets.

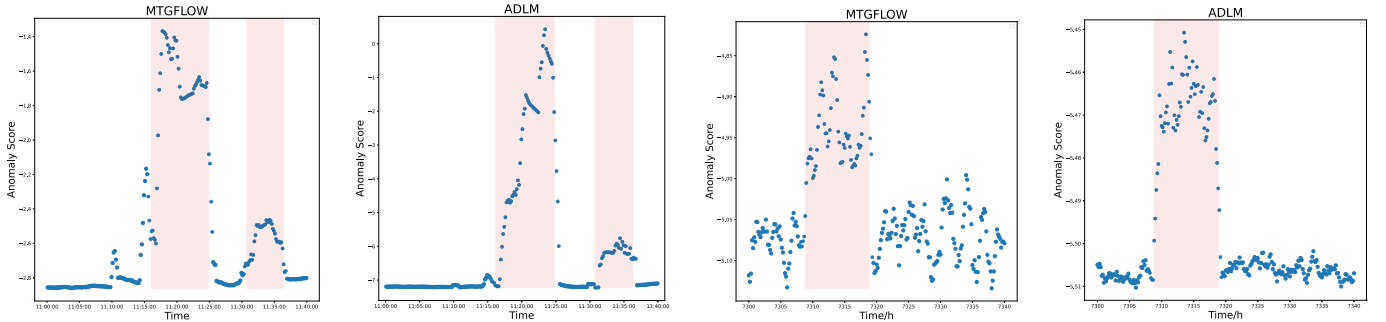


Fig. 6. Point plot of the anomalies prediction of MTGFlow and ALDM on SWaT (left 2 columns) and TEPE (right 2 columns). The x-axis denotes time points, and the y-axis denotes the anomaly score. The red region is ground truth anomalies.

These findings align with the observed performance gains. ALDM delivers larger AUROC improvements on SWaT (+ 5.7%) and MSL (+ 2.9%), i.e., the two most non-stationary datasets, where data drift likely causes greater class ambiguity. On the more stationary datasets (PSM and TEPE), where drift is limited, the gains are smaller (+ 1.1% and + 1.4%, respectively). This supports the hypothesis that ALDM is particularly

effective in scenarios with pronounced non-stationarity and associated data drift.

Next, we analyzed the probability density distributions. Fig. 7 presents probability density plots comparing the distributions of anomaly scores for “Normal” and “Abnormal” data segments. The x-axis represents the anomaly score, while the y-axis shows the corresponding probability density of

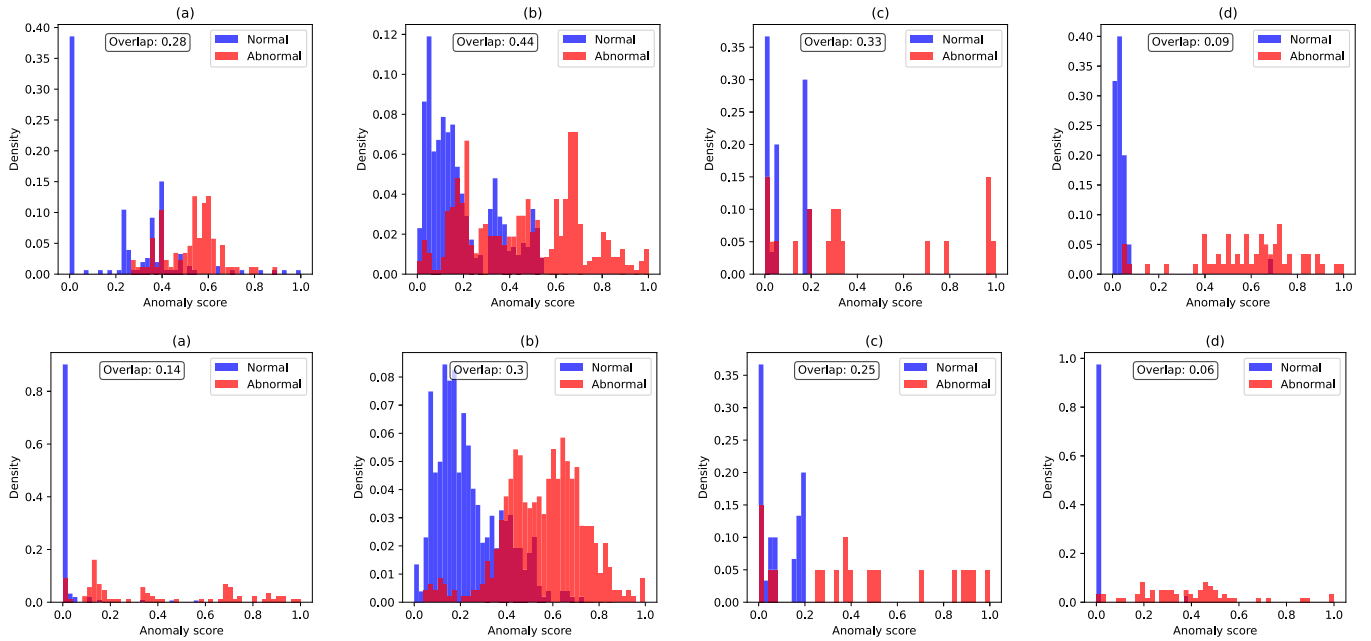


Fig. 7. Probability density plots of MTGFlow (up) and ALDM (bottom) on SWaT (a), PSM (b), MSL (c), and TEPE (d).

TABLE IV

STATIONARITY ASSESSMENT USING AUGMENTED DICKEY-FULLER (ADF) TEST STATISTICS

	SWaT	PSM	MSL	TEPE
ADF Statistics	-0.780	-6.060	-1.228	-9.48
<i>p</i> -value	0.83	0.05	0.66	0.01

segments. As expected, normal data segments are skewed towards lower anomaly scores (left side), while anomalies are skewed towards higher scores (right side). The overlap metric quantifies the area where these two distributions overlap, serving as an indicator of how effectively a method separates normal and abnormal cases. Better separation manifests as less overlap. The first row of subfigures in Fig. 7 displays results achieved by MTGFlow on the four datasets, while the second row shows results from the proposed ALDM.

Regarding the overlap metric: MTGFlow exhibits overlap values of 0.28, 0.44, 0.33, and 0.09 across subfigures (a), (b), (c), and (d), respectively. Crucially, ALDM demonstrates smaller overlap areas than MTGFlow across all datasets. This reduction in overlap indicates that ALDM achieves superior performance in distinguishing between normal and abnormal data, resulting in a clearer boundary.

Regarding the distribution shape: In all subfigures, ALDM's distribution for normal data (blue) exhibits a smaller mean value, while its distribution for abnormal data (red) exhibits a larger mean value compared to MTGFlow. This increased separation between the means further confirms ALDM's enhanced ability to identify anomalies.

Regarding the range of anomaly scores: Compared to MTGFlow, ALDM compresses the range of anomaly scores across all datasets. For instance, considering subfigure (a): MTGFlow assigns normal segments scores ranging from

0 to 1. In contrast, ALDM compresses the anomaly scores of most normal segments tightly around 0.02, demonstrating excellent separation from abnormal segments.

In summary, ALDM shows better performance across all test conditions, especially in reducing the overlap between normal and abnormal data and improving the accuracy of anomaly detection.

E. Ablation Study

To assess the validity of each designed module, we conducted several ablation experiments on the proposed methods.

1) Module Effectiveness: The ablation experiments presented in Table V provide insights into the contributions of context learning (CL), adaptive distance computation (ADC), and soft assignment (SA) to the performance of various models across different datasets. The results indicate that incorporating SA significantly enhances model performance, as evidenced by the substantial improvements when comparing M0 (which lacks all three components) with M1 (which includes only SA). For instance, on the SWaT dataset, the AUROC increases from 73.5 ± 4.3 to 88.5 ± 0.9 , the F1 score rises from 0.3206 to 0.6592, and the $F1_E$ score increases from 0.4267 to 0.9790. Further enhancements are observed when ADC is added (M2), suggesting that both SA and ADC play crucial roles in improving model accuracy. The inclusion of CL (as seen in M3 and M0) also contributes positively, although its impact is less pronounced compared to SA and ADC. The final model, ALDM, which integrates all three components, achieves the highest performance across most datasets, demonstrating that the combination of CL, ADC, and SA is essential for optimal results. This comprehensive analysis underscores the importance of these components in enhancing the model's ability to learn contextual information,

TABLE V
ABLATION EXPERIMENTS ON CONTEXT LEARNING (CL), ADAPTIVE DISTANCE COMPUTATION (ADC), AND SOFT ASSIGNMENT (SA)

Model	CL	ALDP			SWaT			PSM			MSL			TEPE		
		ADC	SA		AUROC	F1	F1 _E	AUROC	F1	F1 _E	AUROC	F1	F1 _E	AUROC	F1	F1 _E
M0	×	×	×		73.5±4.3	0.3206	0.4267	84.8±3.1	0.7804	0.6669	66.3±1.4	0.3911	0.4905	81.2±0.5	0.5855	0.7032
M1	×	×	✓		88.5±0.9	0.6592	0.8790	83.9±1.6	0.7889	0.6770	69.1±1.7	0.3915	0.4929	82.2±1.1	0.6081	0.7387
M2	×	✓	✓		<u>90.0±0.6</u>	<u>0.6786</u>	<u>0.8979</u>	84.9±4.7	0.7864	<u>0.7026</u>	66.7±3.8	<u>0.3925</u>	0.4934	81.4±1.4	0.6120	0.7372
M3	✓	×	×		84.8±1.5	0.4751	0.7702	<u>85.7±1.5</u>	0.7520	0.6743	67.2±1.7	0.3698	0.4949	81.1±0.2	0.6057	0.6750
M4	✓	×	✓		87.9±5.5	0.5769	0.8732	82.6±7.1	0.7748	0.6916	69.1±2.7	0.3910	<u>0.4953</u>	82.1±1.3	0.6244	0.7199
ALDM	✓	✓	✓		90.5±1.3	0.6881	0.8985	86.8±3.4	0.7926	0.7516	70.1±0.8	0.3926	0.5303	82.5±0.8	<u>0.6175</u>	0.7460

TABLE VI
ABLATION EXPERIMENTS ON STATIC DISTANCE METRICS AND ADAPTIVE DISTANCE COMPUTATION (ADC)

	SWaT			PSM			MSL			TEPE		
	AUROC	F1	F1 _E	AUROC	F1	F1 _E	AUROC	F1	F1 _E	AUROC	F1	F1 _E
TD	<u>87.9±5.5</u>	0.5769	<u>0.8732</u>	82.6±7.1	0.7748	0.6916	<u>69.1±2.7</u>	<u>0.3910</u>	<u>0.4953</u>	82.1±1.3	0.6244	0.7199
COS	87.2±3.0	0.6218	0.8016	<u>83.7±6.7</u>	<u>0.7865</u>	<u>0.7312</u>	67.7±2.8	0.3824	0.4929	82.3±0.7	0.6138	<u>0.7309</u>
DTW	87.5±4.6	<u>0.6679</u>	0.7740	83.4±6.7	0.7849	0.7089	68.6±2.8	0.3839	0.4943	81.4±1.9	0.6169	0.7096
ADC [Ours]	90.5±1.3	0.6881	0.8985	86.8±3.4	0.7926	0.7516	70.1±0.8	0.3926	0.5303	82.5±0.8	<u>0.6175</u>	0.7460

compute distances adaptively, and assign labels softly, leading to superior detection performance.

2) *Impact of Distance Metrics*: To further evaluate the benefits of adaptive distance computation, we present a comprehensive comparison between static distance metrics (TD, COS, DTW) and the Adaptive Distance Computation (ADC) method across four datasets in Table VI. The evaluation metrics include AUROC, F1 score, and F1_E score, which collectively offer insights into the models' capabilities in distinguishing normal from anomalous instances, balancing point-wise precision/recall, and addressing event-wise anomaly detection, respectively. ADC consistently outperforms the static distance metrics across nearly every evaluation measure and dataset. These observations suggest that ADC not only provides superior discrimination power but also ensures better precision-recall trade-off and enhanced event-wise detection capability. Furthermore, the relatively smaller standard deviations achieved by ADC indicate its stability and robustness across different configurations. In contrast, while TD, COS, and DTW exhibit varying degrees of effectiveness depending on the dataset, none of them can universally outmatch ADC. This evidence shows the advantage of employing adaptive mechanisms in distance calculation for improving the accuracy and reliability of anomaly detection systems.

We also validated the effectiveness of Cholesky decomposition in ADC by comparing it with two alternatives: eigenvalue clipping and directly learning the metric matrix M . Directly learning M caused immediate training failure due to NaN loss, underscoring the need to enforce positive semi-definiteness. Eigenvalue clipping required costly projections onto the positive semi-definite cone after each iteration, leading to computational overhead (283 seconds/epoch vs. 30 seconds for Cholesky on SWaT) and potential optimization instability.

3) *Impact on Event-Wise Index*: The complete model (ALDM) achieves the best event-level F1_E scores across all four datasets (SWaT, PSM, MSL, TEPE). This demonstrates

that the combined components effectively improve the completeness of anomaly event detection. SA provides significant boost to event-level performance. For instance, on SWaT, adding SA (M0 vs. M1) improves F1_E from 0.4267 to 0.8790. Stable gains are observed on other datasets, indicating SA's crucial role in reducing missed detections for continuous events. Building upon SA, ADC further refines event detection. Comparing M1 (w/o ADC) to M2 (w/ ADC), F1_E improves on SWaT (0.8790 to 0.8979) and PSM (0.6770 to 0.7026). This suggests ADC helps better delineate event boundaries through learned similarity. The impact of CL is dataset-dependent. While it offers a large gain on SWaT (M0 vs. M3: 0.4267 to 0.7702), its effect is more modest or slightly negative on others (e.g., TEPE). This implies its utility is tied to the specific contextual nature of anomalies in the data. The full combination (CL+ADC+SA) yields the best overall event-level performance. Notably, adding ADC to the CL+SA setup (M4 vs. ALDM) leads to substantial gains on PSM (0.6916 to 0.7516) and MSL (0.4953 to 0.5303), confirming ADC's key role in synthesizing context and soft assignment for superior event detection.

4) *Analysis of Soft Assignment*: Based on the ablation studies (Table V), the soft assignment (SA) mechanism delivers largest performance gains. Its most significant impact is observed on the SWaT dataset, where introducing SA alone (M0 vs. M1) boosts AUROC by 15.0 points and F1-score by 0.3386. When integrated with the context learning module (M3 vs. M4), SA provides further improvements, such as increasing the F1-score on SWaT by 0.1018. These quantitative results confirm the importance of SA in enhancing the model's discriminative power. By generating soft labels to guide the latent encoding, SA significantly refines the estimation of data probabilities, leading to more accurate anomaly detection across multiple benchmarks. We also performed a t-SNE visualization of the latent vector following SA-guided latent projection, as shown in Fig. 8. The figure illustrates the distribution of latent vectors at 1, 30, 60, and 100 epochs. It is

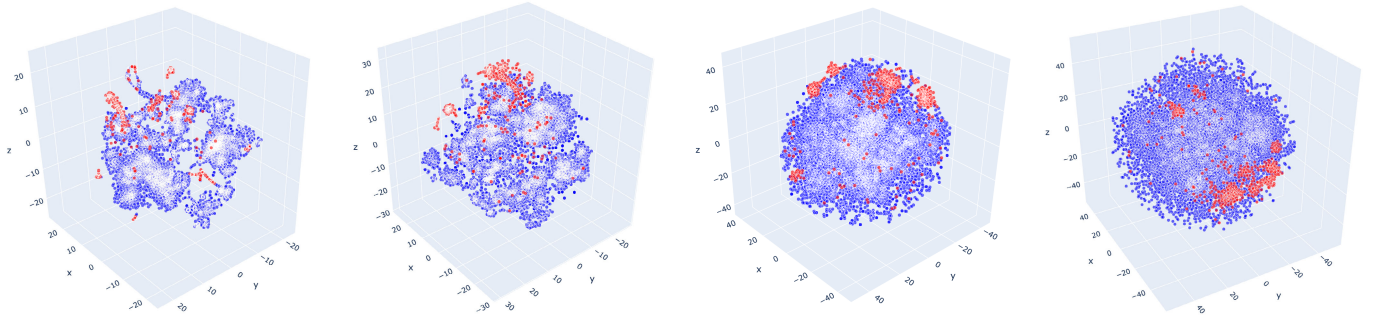


Fig. 8. The t-SNE visualization of the latent vector on SWaT after 1, 30, 60, and 100 epochs, from left to right. The blue and red points correspond to normal and anomalous samples, respectively.

TABLE VII
TRAINING TIME OF ALDM AND BASELINES ON WHOLE
DATASET (MIN)

Model	SWaT	PSM	MSL	TEPE
DeepSVDD	0.23	0.02	0.02	0.21
DeepSAD	0.23	0.03	0.02	0.23
USAD	1.40	0.40	0.50	1.64
DAGMM	1.76	0.35	0.34	1.64
GANF	2.90	0.30	0.61	4.55
FITS	6.41	1.01	1.66	2.30
MTGFlow	2.86	0.30	0.49	4.17
CrossAD	8.25	2.88	3.48	16.58
ALDM [Ours]	38.34	6.74	6.09	18.31

TABLE VIII
TEST TIME OF ALDM AND BASELINES ON WHOLE
DATASET (SEC)

Model	SWaT	PSM	MSL	TEPE
DeepSVDD	3.00	0.17	0.12	1.55
DeepSAD	2.84	0.19	0.13	1.58
USAD	3.96	0.17	0.18	1.67
DAGMM	4.48	1.60	1.66	3.21
GANF	6.64	0.53	1.55	2.17
FITS	34.14	5.98	5.63	42.47
MTGFlow	4.84	1.39	1.40	3.66
CrossAD	7.75	2.16	2.25	15.60
ALDM [Ours]	9.36	2.59	2.46	7.58

evident that the boundary between normal and abnormal samples becomes increasingly clear, indicating that the SA-guided latent projection effectively helps to separate the distributions of normal and abnormal data.

Reviewing the ablation studies presented in Tables V and VI, the following conclusions can be drawn: 1) The proposed ALDM consistently improves performance across experiments. This validates the underlying motivation that increasing the margin between normal data and anomalies through contrastive latent encoding is effective. 2) The best static distance metric varies depending on the dataset, suggesting that different variable dependencies favor different metrics. Crucially, the proposed ADC module addresses this by adaptively learning the optimal metric, enabling ALDM to consistently achieve state-of-the-art (SOTA) performance. 3) The context learning module further enhances ALDM's effectiveness. This demonstrates that context information is valuable not only for modeling data distributions but also for modeling latent distributions.

F. Computational Complexity

The computational complexity of ALDM is approximately 223 MFLOPs (i.e., 2.23×10^8 floating-point operations) per inference, measured in single-precision (FP32) arithmetic. This level of computational demand is well within the capabilities of modern edge devices and mobile platforms. For instance, on the NVIDIA Jetson Nano, a modest embedded platform with 141 GFLOPs, the model is estimated to achieve a latency of 20–50 ms per inference. This satisfies real-time

requirements for most time series monitoring applications. We also compared the training time and inference time of ALDM and other baselines on whole dataset. The results are shown in Tables VII and VIII.

ALDM shows a trade-off between improved performance and increased computation. During training, ALDM takes much longer than all baseline methods on every dataset. For example, on SWaT, ALDM requires 38.34 minutes—about 6 times longer than FITS (6.41 min) and over 13 times longer than the top competitor MTGFlow (2.86 min). Similar gaps appear on PSM (6.74 vs. 1.01 min for FITS), MSL (6.09 vs. 0.61 min for GANF), and TEPE (18.31 vs. 4.55 min for GANF). The longer training time is attributed to the synergistic impact of the latent encoding and contextual learning modules within the ALDM. This collaboration necessitates additional epochs for the model to effectively map data to a suitable latent distribution. Although ALDM requires a longer training duration compared to other benchmark models, the training completion time remains on the minute level (less than an hour), which is still acceptable.

At test time, however, ALDM is more efficient. Though slower than lightweight models like DeepSVDD and USAD, its inference speed is still practical for real-time inference. On SWaT, ALDM processes the full test set in 9.36 seconds—much faster than FITS (74.49s) and only 2–3 times slower than MTGFlow (4.84s). This trend continues on other datasets, with test times of 2.59s (PSM), 2.46s (MSL), and 7.58s (TEPE). The reasonable inference cost, despite the model's complexity, means that once trained, ALDM can compute representations

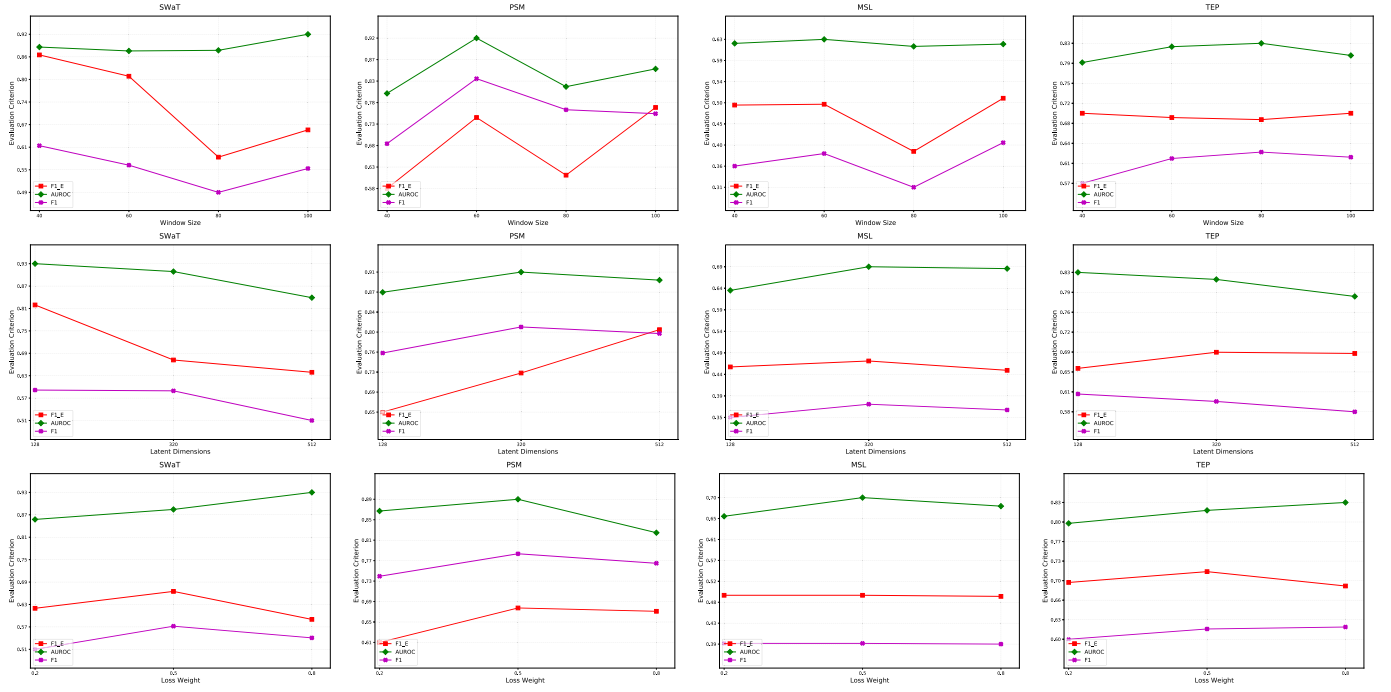


Fig. 9. Sensitivity analysis of window size (up), latent dimension (mid), and loss balance weight (down) in SWaT, PSM, MSL, and TEPE.

and density estimates efficiently. This balance makes ALDM viable for applications where offline training is acceptable and detection accuracy is prioritized.

G. Parameter Sensitivity Analysis

To better understand the robustness of ALDM and facilitate its practical deployment, we conduct sensitivity analysis on three key hyperparameters: window size (T), latent dimension (d), and loss balancing weight (λ). We evaluate their impact across four datasets using AUROC, F1 score, and $F1_E$ score. The experiments are performed in the range of $T \in [40, 100]$, $d \in [120, 512]$ and $\lambda \in [0.2, 0.8]$.

The experimental results are shown in Fig. 9. It can be observed that window size has a significant effect on anomaly detection performance, especially on the SWaT and PSM datasets, where AUROC and $F1_E$ peak when the window size is between 60–80. This indicates that an appropriate context length helps capture anomaly patterns, but overly long windows introduce redundant information, reducing detection accuracy. The impact of latent dimension is relatively smaller, but excessively high dimensions (such as 512) can lead to a slight performance drop, suggesting a potential risk of overfitting. Therefore, it is recommended to use moderate dimensions (such as 256–320). Additionally, the loss balance weight is also critical for model performance. On most datasets, AUROC and $F1_E$ show the best performance when the weight is set to 0.5, indicating that the effective balance between contrastive loss and distribution projection loss is crucial.

These findings underscore that reasonable hyperparameter selection is necessary for high-performance time series anomaly detection. In particular, window size and loss weighting require dataset-specific tuning, whereas latent dimension

can often be fixed to a moderate value without significant performance loss.

H. Limitations

While ALDM demonstrates strong performance in unsupervised anomaly detection, several limitations are worth discussing. First, when abundant and diverse normal data that covers all typical scenarios is available (semi-supervised), reconstruction-based or forecasting-based methods may outperform ALDM’s density estimation approach. These methods explicitly learn the decision boundary of normal data, whereas ALDM does not directly utilize the presence of clean training samples.

Second, the anomaly threshold is determined heuristically via the IQR rule on log-likelihoods. While unsupervised and parameter-free, this approach assumes that the majority of training data is normal (a common but not universal assumption). In scenarios with high contamination rates (>25% anomalies in training data), the IQR-based threshold may be biased, resulting in degraded generalization.

Third, while Normalizing Flows provide exact likelihood estimation, their representational capacity is constrained by the architecture of the invertible transformations. This architectural limitation may yield inadequate approximations of highly multimodal or intricately structured latent distributions, potentially causing false positives/negatives.

These limitations point to promising directions for future work, such as incorporating robust thresholding mechanisms or replacing NFs with more flexible generative priors.

V. CONCLUSION

This study addressed the class ambiguity problem in industrial time series anomaly detection by proposing a novel latent

distribution modeling approach. Our key finding demonstrates that shifting the distribution modeling objective from the raw data space to a latent space effectively enlarges the margin between normal and abnormal patterns, thereby enhancing the discrimination capability of density-based anomaly detectors. Furthermore, the proposed adaptive distance computation mechanism actively separates representations of normal and anomalous segments, making the proposed ALDM well-suited for the anomaly detection task. The efficacy and generality of the proposed latent space transformation principle suggest its potential for adapting and improving existing density-based anomaly detection methods. While this work successfully employed a contrastive method for latent encoding, future research could explore alternative projection techniques to optimize latent space separability and model performance.

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